

Computer Vision -Construction-Site PPE Safety Detection

M4U3 Assignment Mini Report | MAICEN-0526 - Group 5

EXECUTIVE SUMMARY

YOLO11 Nano was evaluated for construction-site PPE and selected safety-condition detection. M1 achieved 96.93% mAP50 and 80.63% mAP50-95. External challenge testing exposed domain-shift weaknesses, motivating M2 targeted augmentation. M2 improved recall from 94.2% to 94.8% and bare-arms test mAP50 from 91% to 93%, while precision and mAP50-95 declined. The result is a controlled performance trade-off rather than a uniformly superior second model.

Rationale for Adaptations to the Reference Model Architecture and Dataset Split

As an applied research/experiment extension to the reference configuration our Group 5, YOLO11 Nano was adopted to explore an alternative contemporary lightweight Ultralytics model architecture, while an 85/9/6 train/validation/test dataset split was used to maintain distinct held-out validation and test sets. These configurations were applied consistently across M1 and M2, preserving controlled experimental comparability, and are explicitly documented to ensure transparency and reproducibility. This extension also provided practical insight into applying an alternative YOLO architecture and dataset-splitting strategy to the same AECO object-detection problem while retaining the core experimental and evaluation framework.

1. Problem, Dataset and Experimental Design

The project investigates whether computer vision can support AECO safety monitoring by identifying workers and selected PPE-related conditions across variable construction imagery. The Roboflow Universe Site Construction Safety dataset contained 3,858 annotated images across seven classes: Person, Helmet, Non-Helmet, Vest, Gloves, Shoes and bare-arms. The preserved split was 3,287 train, 340 validation and 231 test images (85% / 9% / 6%). Images were resized to 640 x 640 using fit with black-edge padding.

Experiment	Model / treatment	Held-out sets	Purpose
M1	YOLO11 Nano; no added augmentation	Original validation/test	Baseline
M2	YOLO11 Nano; 2x training augmentation: flip, crop 0-10%, rotation +/-15°, brightness +/-15%	Same validation/test	Robustness refinement

Table 2. Controlled experimental design

2. Quantitative Results

Metric	M1	M2	Change
Precision	97.1%	95.7%	-1.4 pp
Recall	94.2%	94.8%	+0.6 pp
mAP50	96.93%	96.69%	-0.24 pp
mAP50-95	80.63%	78.58%	-2.05 pp
F1	95.6%	95.2%	-0.4 pp

Table 3. M1 versus M2 headline performance

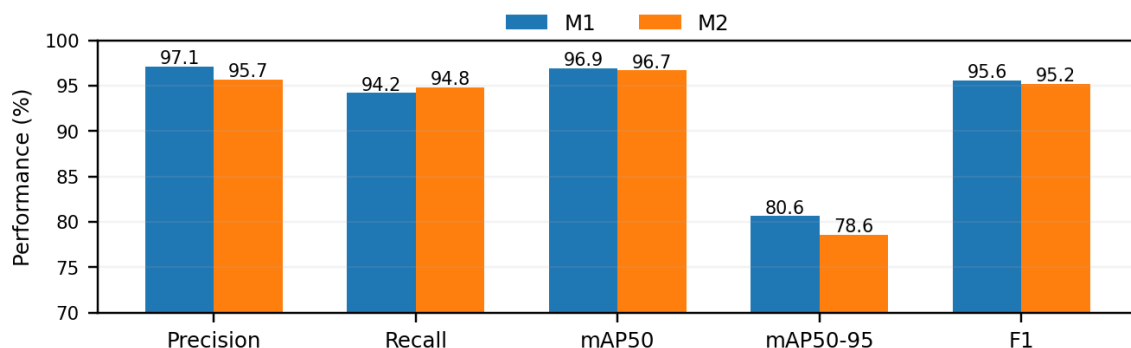


Figure 1. Headline performance comparison - M1 and M2

Interpretation. M2 did not improve every metric. mAP50 was essentially stable (-0.24 pp), recall improved by 0.6 pp and bare-arms test mAP50 improved by 2 pp. Precision fell by 1.4 pp and mAP50-95 by 2.05 pp.

3. External Challenge Testing and Error Analysis

M1 was challenged with genuine external construction/PPE images outside the source dataset using fixed 50% confidence and 50% overlap thresholds. Because these images were not manually annotated as a ground-truth evaluation set, this was qualitative generalisation testing rather than formal external mAP.

Observed failure	Type	Likely cause	Priority response
Visible PPE/safety condition missed	FN	Domain shift / appearance	Add difficult PPE and non-compliance examples
Complex scaffolding scene misses	FN	Occlusion / small scale	Add occluded, small and varied-view examples
Distant/background person missed	FN	Small object / distance	Increase distant-worker coverage

Table 4. Preserved external failure patterns

Additional testing identified weaknesses involving Non-Helmet, bare-arms and Gloves. The ontology contains Non-Helmet but not Non-Vest, Non-Gloves or Non-Shoes; therefore absence of a positive PPE detection must not automatically be interpreted as a violation.

4. Reproducibility and Evidence

- GitHub repository includes README, requirements, Colab-compatible reproducibility notebook, documentation and selected result evidence.
- Notebook verification recorded Python 3.13.15, Ultralytics 8.4.156 and Roboflow 1.5.0; the notebook was restarted and run successfully.
- Historical Roboflow training is preserved as experimental evidence rather than falsely re-executed in Colab. Approximate hosted training time: M1 1 hour; M2 6 hours.
- Evidence includes M1/M2 curves and performance summaries, three annotation examples and five genuine external challenge-test screenshots.

Evidence qualification. The brief requests 10 validation-prediction screenshots and three false-positive examples. A complete defensible set was not preserved. These gaps are disclosed rather than recreated or simulated; held-out metrics, curves, dataset partitions and genuine external evidence remain traceable.

5. Governance, Licensing and Responsible Use

Governance area	Project position
Dataset rights	Roboflow Universe source dataset recorded as CC BY 4.0; preserve attribution.
Model licensing	Roboflow model pages record AGPL-3.0 for the trained models.
Privacy / minimisation	Use authorised site imagery and minimise collection and retention of identifiable worker imagery.
Safety risk	False negatives can miss genuine safety conditions; false positives can create unnecessary alerts. Human review remains necessary.
When not to use	Do not use this proof-of-concept as an autonomous compliance decision system or sole evidence of worker misconduct.

Table 5. Governance and responsible-use controls

6. Prioritised Next Improvements

1. Collect and annotate more representative difficult PPE/non-compliance examples, particularly Non-Helmet, bare-arms and Gloves under external conditions.
2. Improve small/cropped-object robustness using distant workers, partial framing, occlusion and varied viewpoints; assess higher-resolution inference where justified.
3. Strengthen the ontology/rule layer by explicitly representing missing PPE states where required rather than treating absence of a positive detection as a compliance decision.

7. Executive Conclusion

The project achieved strong held-out performance and demonstrated a disciplined cycle of baseline measurement, external challenge testing, error analysis, targeted refinement, controlled comparison and governance review. M2 improved recall and the weakest bare-arms class but did not improve all metrics. The principal next step is better-targeted real-world data and stronger compliance logic, not simply additional training. The model should remain a human-support tool until broader, manually annotated external validation demonstrates reliable deployment performance.